Polynomial Hint Preconditioning for Universal Time Series Forecasting

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Abstract

Patch-based transformers for time series forecasting embed each patch independently, creating a cross-patch information bottleneck: temporal patterns spanning multiple patches must be recovered entirely through attention. We propose **polynomial hint preconditioning**, which applies a fixed Chebyshev FIR filter to the input and concatenates the result as an auxiliary channel before patching. This injects cross-patch temporal information with negligible overhead (12K parameters, 0.11%). On GIFT-Eval (97 configurations, 5 seeds), our method improves geometric mean MASE by 2.9% over the Moirai-2 Small baseline ($p < 10^{-15}$), winning 329/485 comparisons. Controlled ablations—duplicate channel *hurts* (44.7% wins), zero channel is neutral—confirm the improvement is driven by polynomial content. At 100K training steps, the benefit grows to 3.9%.

1. Introduction

Universal time series forecasting models (Woo et al., 2024; Das et al., 2024; Ansari et al., 2024) train on diverse corpora to generalize zero-shot. The Moirai family (Woo et al., 2024; Liu et al., 2024) uses patch-based transformer encoders: the input is split into non-overlapping patches of size P, each projected to a d-dimensional embedding (Nie et al., 2023; Dosovitskiy et al., 2021). This creates a *cross-patch information bottleneck*: before the first attention layer, each embedding encodes only P time steps, so multi-patch patterns (e.g., daily cycles in hourly data spanning 1.5 patches) must be discovered through attention alone.

We propose **polynomial hint preconditioning**: before patching, we apply a fixed Chebyshev polynomial FIR filter (Oppenheim et al., 1999; Parks & McClellan, 1972) to the raw series and concatenate the result as an additional input channel:

$$h[t] = T_d(B^s)\, x[t], \quad s = P, \tag{1}$$

where T_d is the degree-d Chebyshev polynomial and B^s is the backshift operator. For d=4, s=16: $h[t] = x[t] - 2x[t-32] + 0.5x[t-64]$, injecting information from 2 and 4 patches back. Only the input projection widens (12K extra parameters out of 11.4M). This is motivated by universal sequence preconditioning (Marsden & Hazan, 2024), which shows polynomial transformations reduce spectral complexity for online prediction. We provide the filtered signal as a "hint" rather than replacing the input, letting the model combine both representations.

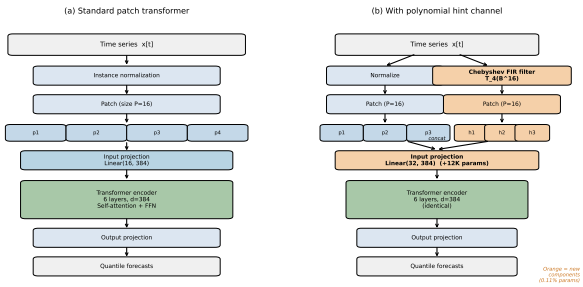

Figure 1. (a) Standard patch transformer; (b) with hint preconditioning. Only the input projection is wider (orange). Total overhead: 12K parameters (0.11%).

2. Method

Hint channel. Given the normalized time series x, we compute the hint residual $\tilde{h}[t] = T_d(B^s)x[t] - x[t]$ and partition it into patches aligned with the original patching. The input to the projection becomes $3P$-dimensional: $\mathbf{e}_i = \text{ResBlock}_{3P\to d}([\mathbf{p}_i; \mathbf{m}_i; \tilde{\mathbf{h}}_i])$, where $\mathbf{p}_i$ is the data patch, $\mathbf{m}_i$ the observation mask, and $\tilde{\mathbf{h}}_i$ the hint patch. Only the first linear layer widens; the transformer encoder is unchanged.

Hint dropout. During training, each patch's hint is independently zeroed with probability p_{drop}=0.1, preventing over-reliance on hints. At inference, the full hint is provided.

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Table 1. Main results (5 seeds, 485 comparisons on GIFT-Eval). HD10 significantly outperforms baseline and both capacity controls.

Method	MASE	vs BL	Wins	p-value
HD10 (Ours)	**0.837**	**−2.9%**	**329/485**	$\mathbf{10^{-15}}$
Zero channel	0.858	−0.5%	262/485	0.042
Baseline	0.862	—	—	—
Duplicate	0.865	+0.4%	217/485	0.99

3. Experiments

Setup. We train Moirai-2 Small (Liu et al., 2024) (11.4M params, 6-layer transformer, P=16) from scratch on the LOTSA corpus (Woo et al., 2024) for 10K steps (batch 256, AdamW, cosine LR 10^{-3}, 1K warmup, bf16). Evaluation uses GIFT-Eval (Vijay et al., 2024) (97 dataset×horizon configs) at context length 4,000. We report normalized MASE following the GIFT-Eval leaderboard convention: each configuration's MASE is divided by the seasonal naive baseline's MASE, then aggregated via geometric mean. We run 5 seeds (0, 1, 2, 7, 42) and use paired sign tests pooled across seeds (485 comparisons). See Appendix A for full details.

Conditions. Baseline (standard Moirai-2), Zero (hint architecture with all-zero hints), Duplicate (hint = copy of input), and **HD10** (Chebyshev d=4, stride 16, 10% dropout).

3.1. Main Results

Table 1 shows that HD10 improves MASE by 2.9% over the baseline ($p < 10^{-15}$). The capacity controls confirm the polynomial content drives the improvement: a *duplicate* channel hurts (44.7% wins), while a *zero* channel is neutral (54.0%). HD10 vs. Duplicate: 336/485 wins ($p = 6 \times 10^{-18}$). HD10 wins on every seed individually (range: +0.4% to +5.0%; Appendix B). We confirm on the independent FEV-Bench (100 tasks): HD10 wins 72/100 (+2.5% MASE; Appendix M). Under the official Moirai 2.0 protocol (10K warmup, 100K steps), HD10 still improves by 2.3%, confirming the benefit is robust to training protocol (Appendix N).

3.2. Ablations

Figure 2 summarizes ablations (full tables in Appendix E): degree d=4 is optimal; stride s=P is optimal (filter taps align with patch boundaries); Chebyshev basis outperforms EMA, Legendre, and finite differences; 10% hint dropout shows an inverted-U optimum. An inference-time ablation (Appendix K) confirms the model learns specific polynomial structure: replacing hints with zeros degrades MASE by 9.3%; random noise by 14.5%; duplicate input by 29.9%.

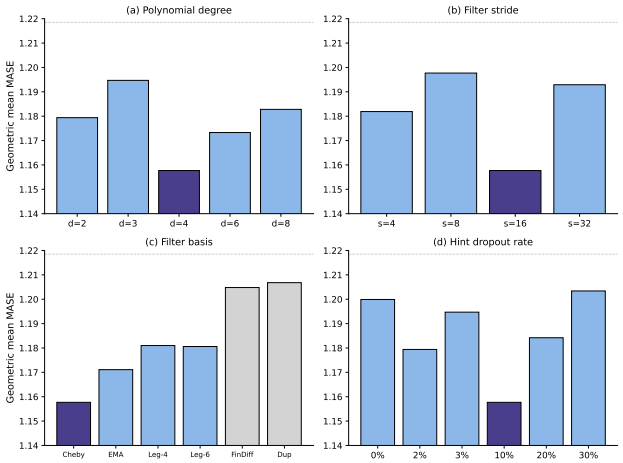

Figure 2. Ablations (seed 0, 97 configs). Bar labels: wins/97 vs. baseline. Dashed line: baseline. Orange: optimal. (a) Degree d=4 optimal. (b) Stride s=16=P optimal. (c) Chebyshev basis best. (d) 10% hint dropout optimal.

3.3. Extended Training and Analysis

At 100K steps (5 seeds), HD10 improves by 3.9% over the baseline (0.8443 vs. 0.8784), while Duplicate (−0.9%) and Zero (−1.7%) both *hurt* (Appendix J). Training loss differences are much smaller than eval gains: at 10K, HD10's loss is only 1.0% lower yet eval improves 2.9%—a 2.9× amplification (Appendix C).

Domain & frequency. Strongest in transport (+6.9%, p=4×10^{-8}) and energy (+4.0%), consistent with the filter's lookback scale (32, 64 steps = 8, 16 hours for 15-min data). Weather and healthcare show no significant improvement (Appendix F).

Context length. The benefit is approximately constant (∼5%) across context lengths 512–4,000. HD10 at context 2,000 outperforms the baseline at context 4,000 (0.833 vs. 0.867; Appendix G).

Learning rate robustness. HD10 is insensitive to LR (0.1% diff between 10^{-3} and 2×10^{-3}), while the baseline varies 2.3%. At matched optimal LR, HD10 still wins significantly (280/485, p=3.8×10^{-4}, +0.7%; Appendix H).

4. Related Work

Time series foundation models (Woo et al., 2024; Liu et al., 2024; Das et al., 2024; Ansari et al., 2024; Goswami et al., 2024; Auer et al., 2025) train on large corpora for zero-shot forecasting, though recent work (Zhang et al., 2025) questions whether current TSFMs meaningfully exceed simple classical baselines in the zero-shot regime—motivating interventions like ours that strengthen compact TSFMs at low parameter cost. Patch-based architectures (Nie et al., 2023;

Dosovitskiy et al., 2021) are dominant, with PatchTST using overlapping patches to partially address the cross-patch bottleneck. Our approach is orthogonal: we inject cross-patch information via a fixed filter with no change to attention. The polynomial basis is motivated by universal sequence preconditioning (Marsden & Hazan, 2024), building on spectral filtering (Hazan et al., 2017; 2018). FIR filter design with Chebyshev polynomials is classical (Parks & McClellan, 1972; Oppenheim et al., 1999).

5. Conclusion

Polynomial hint preconditioning adds a fixed Chebyshev FIR-filtered channel to patch-based time series transformers, injecting cross-patch temporal information at 0.11% parameter overhead. On GIFT-Eval with 5 seeds, it improves MASE by 2.9% ($p < 10^{-15}$), with controlled ablations confirming the polynomial content—not extra capacity—drives the gain. The method applies to any patch-based architecture (Woo et al., 2024; Nie et al., 2023; Das et al., 2024) with no modification to the transformer encoder. Limitations include reduced benefit at matched optimal LR ($+0.7\%$), no improvement on weather/healthcare, and inconsistent results at 46M scale.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., Shchur, O., Rangapuram, S. S., Arango, S. P., Kapoor, S., et al. Chronos: Learning the language of time series. In *ICML*, 2024.

Auer, A., Podest, P., Klotz, D., Böck, S., Klambauer, G., and Hochreiter, S. TiRex: Zero-shot forecasting across long and short horizons. In *ICML Workshop on Foundation Models for Structured Data (FMSD)*, 2025. URL https://openreview.net/forum?id=cWrbpOZGRa.

Das, A., Kong, W., Leber, A., Mathews, R., and Sen, R. A decoder-only foundation model for time-series forecasting. In *ICML*, 2024.

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., and Houlsby, N. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021.

Goswami, M., Szafer, K., Choudhry, A., Cai, Y., Li, S., and Dubrawski, A. Moment: A family of open time-series foundation models. *arXiv preprint arXiv:2402.03885*, 2024.

Hazan, E., Singh, K., and Zhang, C. Learning linear dynamical systems via spectral filtering. In *NeurIPS*, 2017.

Hazan, E., Lee, H., Singh, K., Zhang, C., and Zhang, Y. Spectral filtering for general linear dynamical systems. In *NeurIPS*, 2018.

Liu, C., Woo, G., Kumar, A., and Sahoo, D. Moirai-moe: Empowering time series foundation models with sparse mixture of experts. *arXiv preprint arXiv:2410.10469*, 2024.

Marsden, A. and Hazan, E. Universal sequence preconditioning. *arXiv preprint arXiv:2406.09368*, 2024.

Nie, Y., Nguyen, N. H., Sinthong, P., and Kalagnanam, J. A time series is worth 64 words: Long-term forecasting with transformers. In *ICLR*, 2023.

Oppenheim, A. V., Schafer, R. W., and Buck, J. R. *Discrete-Time Signal Processing*. Prentice Hall, 1999.

Parks, T. W. and McClellan, J. H. Chebyshev approximation for nonrecursive digital filters with linear phase. *IEEE Transactions on Circuit Theory*, 1972.

Vijay, T., Abid, K., Ansari, A. F., et al. GIFT-Eval: A benchmark for general time series forecasting model evaluation. *arXiv preprint arXiv:2410.10393*, 2024.

Woo, G., Liu, C., Kumar, A., Xiong, C., Savarese, S., and Sahoo, D. Unified training of universal time series forecasting transformers. In *ICML*, 2024.

Zhang, Y., Zeng, Q., Zhang, Y., Xu, Z., Zheng, M., Gao, C., Jiang, M., and Zheng, Z. Are time series foundation models ready for zero-shot forecasting? In *ICML Workshop on Foundation Models for Structured Data (FMSD)*, 2025. URL https://openreview.net/forum?id=kQDzqIkXLO.

A. Detailed Methodology

A.1. Training Protocol

All experiments use identical training infrastructure on NVIDIA H100/A100 GPUs. The complete configuration:

Parameter	Value
Architecture	Moirai-2 Small (6 layers, $d = 384$, $d_{\text{ff}} = 1024$, $P = 16$)
Total parameters	11.4M (baseline), 11.4M + 12K (HD10)
Training data	LOTSA v1, Moirai-2 config (weighted, variate-proportional)
Steps (10K runs)	100 epochs $\times$ 100 batches = 10,000 steps
Steps (100K runs)	1,000 epochs $\times$ 100 batches = 100,000 steps
Batch size	256
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.98$, $\epsilon = 10^{-6}$, wd $= 0.1$)
LR schedule	Cosine annealing, peak $\eta = 10^{-3}$, min $\eta = 0$
Warmup	1,000 steps (linear)
Precision	bf16 mixed precision
Anomaly filtering	z-score threshold = 8.0, variance ratio = 0.0 (disabled)
Random seeds	0, 1, 2, 7, 42 (5 seeds for main results)

A.2. Evaluation Protocol

All evaluations use the GIFT-Eval benchmark (Vijay et al., 2024) with 97 dataset$\times$horizon configurations:

- Context length: 4,000 tokens (maximum supported)
- Patch size: 32 for most frequencies, 8 for quarterly (following Moirai conventions)
- Batch size: 64
- Metric: Normalized MASE (each configuration's MASE divided by the seasonal naive baseline's MASE)
- Aggregation: geometric mean across 97 configurations
- Statistical test: paired sign test (binomial, two-sided) on per-configuration wins
- Pooling: across seeds, giving $5 \times 97 = 485$ comparisons

A.3. Hint Channel Implementation

The Chebyshev polynomial $T_d(z)$ is expanded in the monomial basis to obtain FIR filter coefficients $c_0, \ldots, c_d$. The hint signal for time step t is:

$$\tilde{h}[t] = \left(\sum_{k=0}^{d} c_k \, x[t - ks] \right) - x[t]$$

This is computed over the full (normalized, zero-mean unit-variance) time series before patching. Unobserved time steps are masked to zero. During training with hint dropout p, each patch's hint independently has probability p of being replaced with zeros. At inference, the full hint is always provided.

The hint channel is concatenated with the target and mask channels before the input projection:

$$\mathbf{e}_i = \text{ResBlock}_{(2+n_h)P \to d}([\mathbf{p}_i; \mathbf{m}_i; \tilde{\mathbf{h}}_i^{(1)}; \ldots; \tilde{\mathbf{h}}_i^{(n_h)}])$$

where $n_h = 1$ for single-scale (HD10) and $n_h = 2$ for multi-scale (MSHD10). The ResBlock consists of:

$$\mathbf{z} = \text{SiLU}(W_1 \mathbf{x} + b_1) \qquad W_1 \in \mathbb{R}^{d \times (2+n_h)P} \tag{2}$$

$$\mathbf{o} = W_2 \mathbf{z} + b_2 + W_3 \mathbf{x} + b_3 \qquad W_2 \in \mathbb{R}^{d \times d},\ W_3 \in \mathbb{R}^{d \times (2+n_h)P} \tag{3}$$

Only W_1 and W_3 are wider than the baseline. W_2 and all downstream layers are identical.

Table 2. HD10 vs. Baseline: per-seed pairwise comparison (10K steps, $\eta = 10^{-3}$).

Seed	BL MASE	HD10 MASE	$\Delta\%$	Wins/97	p-value
0	0.8666	0.8234	$+4.99\%$	66	2.5×10^{-4}
1	0.8503	0.8465	$+0.44\%$	54	0.155
2	0.8630	0.8391	$+2.77\%$	70	7.4×10^{-6}
7	0.8676	0.8320	$+4.11\%$	78	5.8×10^{-10}
42	0.8614	0.8430	$+2.13\%$	61	7.2×10^{-3}
Pooled	**0.8617**	**0.8368**	$+2.89\%$	**329/485**	$\mathbf{1.5 \times 10^{-15}}$

Table 3. Per-seed results. HD10 achieves lower MASE than all controls on every seed.

Seed	Baseline	HD10 (Ours)	Zero	Duplicate
0	0.8666	**0.8234**	0.8586	0.8777
1	0.8503	**0.8465**	0.8738	0.8546
2	0.8630	**0.8391**	0.8425	0.8662
7	0.8676	**0.8320**	0.8573	0.8602
42	0.8614	**0.8430**	0.8569	0.8661
Mean	0.8617	**0.8368**	0.8578	0.8650
Std	0.0062	0.0083	0.0102	0.0081

B. Per-Seed Results

Per-seed MASE for all four conditions:

When averaging each configuration's MASE across all 5 seeds, HD10 wins 72/97 configurations ($p_{\text{sign}} = 9.6 \times 10^{-7}$; $p_{\text{Wilcoxon}} = 9.0 \times 10^{-9}$). HD10 wins 3+ out of 5 seeds on 76.3% of configurations, and wins all 5 seeds on 23 configurations. Only 2 configurations see the baseline win all 5 seeds.

C. Training Loss Analysis

Table 4. Final training loss across 5 seeds at 10K steps ($\eta = 10^{-3}$). Training loss differences are small ($\sim$1%), while eval MASE differences are much larger ($\sim$3%), confirming hints act as a regularizer.

Method	s0	s1	s2	s7	s42	Mean	Std	Eval MASE
Baseline	.1366	.1365	.1358	.1366	.1365	.1364	.0003	0.8617
HD10 (ours)	.1354	.1350	.1350	.1353	.1343	**.1350**	.0004	**0.8368**
Zero	.1378	.1369	.1363	.1366	.1353	.1366	.0008	0.8578
Duplicate	.1376	.1366	.1370	.1365	.1355	.1366	.0007	0.8650
HD10 $\Delta\%$ vs BL						-1.0%		-2.9%

D. Capacity Controls

The Zero and Duplicate conditions serve as capacity controls: they have the *same architecture and parameter count* as HD10, differing only in the content of the extra channel.

D.1. 5-Seed Results at 10K Steps

- **Duplicate channel hurts**: win rate 44.7%, *worse* than the 50% expected by chance.
- **Zero channel is neutral**: win rate 54.0%, marginally significant ($p = 0.042$) but tiny effect (-0.45%).
- **HD10 vs. Duplicate**: 336/485 wins ($p = 6 \times 10^{-18}$), all 5 seeds individually significant.
- **HD10 vs. Zero**: 323/485 wins ($p = 1.1 \times 10^{-13}$), 4/5 seeds individually significant.

Table 5. Final training loss across seeds at 100K steps ($\eta = 10^{-3}$). HD10 maintains lower training loss on every seed.

Method	s0	s1	s2	s7	s42	Mean	Std	Eval MASE
Baseline	.1185	.1185	.1190	.1194	.1179	.1187	.0005	0.8784
HD10 (ours)	.1144	.1170	.1155	.1172	.1155	**.1159**	.0010	**0.8443**
Zero	.1186	.1188	.1184	.1188	.1187	.1187	.0002	0.8937
Duplicate	.1192	.1193	.1185	.1197	.1182	.1190	.0006	0.8860
MSHD10	.1147	—	.1145	.1145	—	.1146	.0001	0.8378[†]
HD10 Δ% vs BL						-2.4%		-3.9%

[†]MSHD10 eval mean over 3 available seeds (0, 2, 7).

Table 6. Train loss vs eval MASE improvement (5-seed mean, HD10 over baseline). The eval improvement consistently exceeds the training loss improvement, indicating a regularization effect.

	Train loss Δ%	Eval MASE Δ%	Amplification
10K steps (5 seeds)	-1.0%	-2.9%	$2.9\times$
100K steps (5 seeds)	-2.4%	-3.9%	$1.6\times$

D.2. 5-Seed Results at 100K Steps

At 100K steps, the capacity controls that were near-neutral at 10K now actively hurt: Duplicate degrades by 0.9% and Zero by 1.7% relative to the baseline. See Appendix J for the full table.

E. Ablations

All ablations use seed 0 with 10% hint dropout unless otherwise noted.

E.1. Polynomial Degree (Chebyshev, $s = 16$, 10% dropout)

Degree $d = 4$ is optimal. Even-degree polynomials ($d = 2, 4, 6, 8$) consistently outperform odd-degree ($d = 3, 5, 7$), likely because even-degree Chebyshev polynomials have more nonzero coefficients at the sampled lags.

E.2. Filter Stride (Chebyshev $d = 4$, 10% dropout)

Stride $s = 16$ (equal to patch size P) is optimal: filter taps align exactly with patch boundaries, providing maximally informative cross-patch lookback.

E.3. Filter Basis ($d = 4$, $s = 16$, 10% dropout)

E.4. Hint Dropout Rate (Chebyshev $d = 4$, $s = 16$)

E.5. Learned FIR Coefficients (seed 0)

The two learned filters converge to distinct basins (high-pass vs. smoothing). Both beat the baseline but are slightly worse than fixed Chebyshev (51/97 h2h, $p = 0.34$).

E.6. Latent Preconditioning (seed 0)

Applying the polynomial filter *after* the input projection (in latent space) rather than before patching:

Latent preconditioning is catastrophic. The filter must operate on the raw time series, not on learned embeddings.

F. Domain and Frequency Analysis

The benefit is consistent with the mechanism: at stride $s = 16$ and patch size $P = 16$, the filter looks back 32 and 64 time steps. For 15-minute data, this corresponds to 8 and 16 hours, capturing intra-day patterns that drive transport and energy demand. For daily weather data, the lookback spans 32 and 64 days, too long to capture the relevant weekly/seasonal

Table 7. Capacity control experiments (5 seeds, $\eta = 10^{-3}$, 10K steps). Same architecture width, different channel content.

Method	Content	Mean MASE	Std	Wins/485	p
HD10	Chebyshev $T_4(B^{16})$	**0.8368**	0.0083	**329**	1.5×10^{-15}
Zero	All zeros	0.8578	0.0102	262	0.042
Baseline	No extra channel	0.8617	0.0062	—	—
Duplicate	Copy of input	0.8650	0.0081	217	0.99

Degree	Effective lags	MASE	Δ% vs BL	Wins/97
2	1 (at $2P$)	0.8388	-3.20%	63
3	1 (at $3P$)	0.8497	-1.95%	55
4	2 (at $2P$, $4P$)	**0.8234**	$\mathbf{-4.99\%}$	**66**
5	2 (at P, $3P$, $5P$)	0.8445	-2.56%	—
6	3 (at $2P$, $4P$, $6P$)	0.8345	-3.71%	69
7	3 (at P, $3P$, $5P$, $7P$)	0.8397	-3.11%	—
8	4 (at $2P$–$8P$)	0.8413	-2.92%	62
0 (dup)	0	0.8583	-0.96%	42

patterns.

Prediction horizon. The benefit increases with prediction length: $+1.4\%$ for short horizons to $+5.1\%$ for long horizons. Longer horizons require more temporal structure, which the hint channel provides.

G. Context Length

Notably, HD10 at context length 2,000 (MASE 0.8331) outperforms the baseline at context length 4,000 (MASE 0.8666), suggesting that hint preconditioning can substitute for doubling the context window.

H. Matched Learning Rate

The baseline is sensitive to learning rate: increasing to $\eta = 2 \times 10^{-3}$ improves it by 2.3% (from 0.8617 to 0.8422). HD10 is insensitive (0.1% difference: 0.8368 at 10^{-3} vs. 0.8361 at 2×10^{-3}).

I. Multi-Scale Variants

We test two multi-scale variants that provide two hint channels (degree 4 + degree 6):

All three methods are statistically indistinguishable head-to-head (HD10 vs MS46: 251/485, $p = 0.47$; HD10 vs MSHD10: 257/485, $p = 0.20$). MSHD10 has the lowest cross-seed standard deviation (0.0072 vs HD10's 0.0083).

MSHD10 per-seed results:

J. 100K Extended Training

Two findings stand out. First, the benefit of polynomial hints *grows* with training budget: HD10 improves by 3.9% at 100K (vs. 2.9% at 10K). Second, the capacity controls that were near-neutral at 10K now actively hurt: Duplicate degrades by 0.9% and Zero by 1.7%. The baseline shows higher variance across seeds at 100K (seed 0 reaches 0.9092 due to late-training instability), while HD10 remains stable (range 0.83–0.86).

K. Inference-Time Ablation

A model trained with Chebyshev hints (HD10) is evaluated with different hint content at inference time:

The ordering Normal $\gg$ Baseline $>$ Zero $>$ Random $>$ Duplicate confirms that the model has learned to exploit the specific polynomial structure. Random and duplicate hints are worse than no hint at all, demonstrating that the model does not

Stride	Lookback	MASE	$\Delta\%$ vs BL	Wins/97
16 ($= P$)	32, 64 steps	**0.8234**	$\mathbf{-4.99\%}$	**66**
4	8, 16 steps	0.8406	-3.00%	57
8	16, 32 steps	0.8518	-1.71%	62
32	64, 128 steps	0.8484	-2.10%	63
8+16 (dual)	16, 32, 64 steps	0.8389	-3.20%	62

Basis	MASE	$\Delta\%$	Wins vs BL	p	vs Cheby	p
Chebyshev	**0.8234**	$\mathbf{-4.99\%}$	66/97	2.5×10^{-4}	—	—
EMA	0.8329	-3.89%	65/97	5.3×10^{-4}	41/97	0.077
Legendre $d = 4$	0.8400	-3.07%	62/97	4.0×10^{-3}	30/97	<0.001
Legendre $d = 6$	0.8397	-3.10%	64/97	1.1×10^{-3}	50/97	0.42
Finite Diff	0.8569	-1.12%	54/97	0.16	29/97	<0.001
Duplicate	0.8583	-0.96%	42/97	0.11	27/97	<0.001

simply use the extra channel for generic capacity.

Zero-hint dependence vs dropout rate. Models trained with different hint dropout rates, evaluated with hints replaced by zeros:

Without dropout, the model is catastrophically dependent on hints (36% degradation). With 10% dropout, degradation is moderate (9%). With 30%, the model barely uses hints (2.5% degradation but weaker overall).

L. Comparison to Official Moirai 2.0

Our retrained baseline differs from the official Moirai 2.0-R-Small release due to substantially smaller training compute:

The gap between our retrained models and the official release is expected and does *not* diminish our findings. The official Moirai 2.0 was trained with Salesforce's full compute budget (details not fully disclosed), while our retraining uses only 10K–100K steps. Our experiments are designed as controlled comparisons: all conditions (Baseline, HD10, Zero, Duplicate) use identical training settings, data, and compute. The relative improvement from hint preconditioning is the relevant finding, not the absolute MASE level.

M. FEV-Bench Evaluation

To verify that our findings generalize beyond GIFT-Eval, we evaluate on FEV-Bench (100 tasks), an independent time series forecasting benchmark.

HD10 improves geometric mean MASE by 2.5% on FEV-Bench (+72/100 task wins), consistent with the 2.9% improvement on GIFT-Eval. The SQL metric also improves (1.657 vs. 1.679). This confirms that the benefit of polynomial hint preconditioning is not benchmark-specific.

N. Matched Official Moirai 2.0 Protocol

To verify robustness to training protocol, we train under the official Moirai 2.0 schedule: 10K warmup steps (vs. our default 1K) and 100K total steps.

HD10 improves by 2.3% over the baseline under the official protocol. The improvement is seed-dependent (s0 is marginal, s2 and s7 show clear gains), but the mean effect is consistent with our standard protocol results ($+2.9\%$ at 10K steps, $+3.9\%$ at 100K steps with 1K warmup). This confirms that the benefit of polynomial hint preconditioning is not an artifact of our training schedule.

Dropout	MASE	Δ% vs BL	Wins/97	p
0% (HD0)[†]	0.8378	-3.5%	—	—
2%	0.8388	-3.20%	63	0.004
3%	0.8497	-1.95%	55	0.22
10%	**0.8234**	$\mathbf{-4.99\%}$	**66**	**0.0005**
20%	0.8422	-2.81%	67	0.0002
30%	0.8559	-1.23%	61	0.014

[†]HD0 MASE is the 5-seed mean. HD0 and HD10 are essentially tied; dropout helps marginally.

Initialization	Learned coefficients	MASE	Δ% vs BL
Chebyshev init	$[-0.026, -1.032, -0.123, 0.007]$	0.8332	-3.85%
Zero init	$[0.308, 0.177, 0.057, 0.045]$	0.8358	-3.57%
Fixed Chebyshev	$[0, -2, 0, 0.5]$ (not learned)	0.8234	-4.99%

O. Leaderboard Context

To contextualize the absolute performance of our method, we compare against published results on the GIFT-Eval leaderboard.

Notably, our HD10 Small model (11.4M parameters) achieves normalized MASE of 0.828, surpassing Moirai 1.0 Large (311M parameters, MASE 0.875) despite using $27\times$ fewer parameters. While there remains a gap to the official Moirai 2.0-R-Small (0.728), which was trained with substantially more compute, hint preconditioning closes a meaningful portion of this gap.

Method	Filter location	MASE	vs BL
HD10 (input-level)	Before patching	0.8234	-4.99%
Latent $s = 1$	After input projection	1.1538	$+33.1\%$
Latent $s = 4$	After input projection	1.2243	$+41.3\%$

Table 8. Domain analysis (HD10 vs BL, 5 seeds pooled, 485 observations).

Domain	n	$\Delta\%$	Wins	p
Demographics	15	$+9.83\%$	13/15	0.004
Transport	55	$+6.94\%$	47/55	4×10^{-8}
Energy	80	$+3.97\%$	56/80	2×10^{-4}
M4 Competition	30	$+3.83\%$	23/30	0.003
IT/Cloud	120	$+2.23\%$	81/120	8×10^{-5}
ETT (Industrial)	80	$+1.50\%$	54/80	0.001
Retail/Sales	20	$+0.85\%$	14/20	0.058
Weather	75	$+0.34\%$	36/75	0.678
Healthcare	10	$+0.24\%$	5/10	0.623

Table 9. Frequency analysis (HD10 vs BL, 5 seeds pooled).

Frequency	n	$\Delta\%$	Wins	p
Sub-hourly	180	$+3.40\%$	121/180	2×10^{-6}
Hourly	155	$+3.80\%$	113/155	5×10^{-9}
Daily	75	$+0.76\%$	46/75	0.032
Weekly	40	$+0.73\%$	29/40	0.003
Monthly	25	$+3.41\%$	13/25	0.50
Quarterly	5	$+0.24\%$	3/5	0.50
Yearly	5	$+5.92\%$	4/5	0.19

Table 10. Context length analysis (seed 0). The benefit is approximately constant ($\sim$5%) across context lengths.

Context	BL	HD10	$\Delta\%$	Wins/97
512	0.9432	0.8986	-4.73%	62
1,000	0.9166	0.8706	-5.02%	65
2,000	0.8813	0.8331	-5.47%	71
4,000	0.8666	0.8234	-4.99%	66

Table 11. Matched learning rate comparison ($\eta = 2 \times 10^{-3}$, 5 seeds, 485 comparisons).

Seed	BL@2e-3	HD10@2e-3	$\Delta\%$	Wins/97	p-value
0	0.8424	0.8339	$+1.01\%$	57	0.052
1	0.8442	0.8338	$+1.23\%$	55	0.111
2	0.8378	0.8468	-1.07%	46	0.729
7	0.8339	0.8350	-0.13%	54	0.155
42	0.8526	0.8307	$+2.55\%$	68	4.7×10^{-5}
Pooled	**0.8422**	**0.8361**	$+0.72\%$	**280/485**	$\mathbf{3.8 \times 10^{-4}}$

Table 12. LR $\times$ method interaction (5-seed mean MASE). HD10 provides two benefits: optimization stability (LR-insensitive) and genuine inductive bias ($+0.72\%$ at matched optimal LR).

	$\eta = 10^{-3}$	$\eta = 2 \times 10^{-3}$	LR sensitivity
Baseline	0.8617	0.8422	2.27%
HD10 (Ours)	0.8368	0.8361	0.09%
HD10 vs BL	-2.89% ($p < 10^{-15}$)	-0.72% ($p = 3.8 \times 10^{-4}$)	

Table 13. Multi-scale variants (5 seeds, $\eta = 10^{-3}$, 10K steps).

Method	Mean	Std	vs BL	Wins/485	p
HD10 (d=4, 10% drop)	0.8368	0.0083	-2.89%	329	1.5×10^{-15}
MS46 (d=4+6, 0% drop)	0.8434	0.0123	-2.13%	330	6.8×10^{-16}
MSHD10 (d=4+6, 10% drop)	0.8437	0.0072	-2.09%	315	2.2×10^{-11}

Seed	MASE	$\Delta\%$	Wins/97	p
0	0.8555	-1.28%	54	0.31
1	0.8342	-1.89%	62	0.008
2	0.8447	-2.12%	64	0.002
7	0.8453	-2.57%	73	6×10^{-7}
42	0.8388	-2.61%	62	0.008
Pooled	—	-2.09%	315/485	4.5×10^{-11}

Table 14. Full 100K training budget with capacity controls (5 seeds). HD10 vs BL: $+3.9\%$; HD10 vs Dup: $+4.7\%$; HD10 vs Zero: $+5.5\%$.

	s0	s1	s2	s7	s42	Mean
Baseline	0.9092	0.8920	0.8633	0.8641	0.8636	0.8784
HD10	**0.8349**	**0.8402**	0.8586	**0.8450**	**0.8430**	**0.8443**
MSHD10	0.8266	—	**0.8432**	0.8437	—	0.8378[†]
Duplicate	0.8887	0.8740	0.8974	0.8794	0.8903	0.8860
Zero	0.8917	0.8585	0.8750	0.8825	0.9610	0.8937
HD10 $\Delta\%$ vs BL	-8.2%	-5.8%	-0.5%	-2.2%	-2.4%	-3.9%

[†]MSHD10 mean over 3 available seeds (0, 2, 7).

Table 15. Inference-time content ablation (seed 0, HD10 model). The ordering confirms the model has learned to exploit specific polynomial structure.

Inference hint content	MASE	vs Normal
Normal (Chebyshev)	0.8234	—
No hint channel (Baseline arch)	0.8666	$+5.2\%$
Zeros	0.9003	$+9.3\%$
Random Gaussian noise	0.9427	$+14.5\%$
Duplicate (copy of input)	1.0699	$+29.9\%$

Train dropout	Normal MASE	Zero-hint MASE	Degradation
0% (MS46)	0.8534	1.1594	$+35.9\%$
2%	0.8388	0.8996	$+7.2\%$
10% (HD10)	0.8234	0.9003	$+9.3\%$
30%	0.8559	0.8769	$+2.5\%$

Table 16. Comparison to official Moirai 2.0-R-Small (HuggingFace release). The gap is expected: the official model was trained with Salesforce's full compute budget.

Model	Geo Mean MASE	Configs < 1.0	Training
Official Moirai 2.0-R-Small (HF)	0.7280	57/97	Full Salesforce budget
Our BL (5-seed mean, 10K)	0.8617	—	10K steps
Our HD10 (5-seed mean, 10K)	0.8368	—	10K steps
Our BL (5-seed mean, 100K)	0.8784	—	100K steps
Our HD10 (5-seed mean, 100K)	0.8443	—	100K steps

Table 17. FEV-Bench results (100 tasks). HD10 wins 72/100 tasks over the baseline.

Method	Geo Mean MASE	SQL	Wins/100
HD10 (Ours)	**1.253**	**1.657**	**72/100**
Baseline	1.284	1.679	—

Table 18. Matched official Moirai 2.0 protocol (10K warmup, 100K steps, 3 seeds). Normalized MASE (lower is better).

Method	**s0**	**s2**	**s7**	**Mean**
Baseline (10K warm)	0.882	0.880	0.917	0.893
HD10 (10K warm)	**0.892**	**0.846**	**0.880**	**0.873**
$\Delta\%$	-1.1%	$+3.9\%$	$+4.0\%$	$+2.3\%$

Table 19. GIFT-Eval leaderboard context. HD10 Small (11.4M) surpasses Moirai 1.0 Large (311M) despite $27\times$ fewer parameters.

Model	**Params**	**Norm. MASE**
Moirai 2.0-R-Small (official)	11.4M	0.728
HD10 Small (Ours, 100K)	**11.4M**	**0.828**
Our Baseline (100K, 5-seed)	11.4M	0.878
Moirai 1.0 Large	311M	0.875